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How Do VLMs Behave When Blind or Misled? Behavioral Evaluation of VLMs on Scientific Figures

SciFigBench Reveals That VLMs Confabulate Data Values When Figures Are Blurred and Defer to Incorrect Captions Over Visual Evidence

By Paul Osemudiamé Oamen, Owusu-Banahene Osei, Ananya Mukherjee, Christian Greisinger, Steffen Eger, Pius Onobhayedo, Wei Zhao

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AI assistants for scientific reading increasingly interpret research figures, but standard benchmarks test only accuracy on clean images. SciFigBench stress-tests VLMs on 250 scientific figures across 34,000+ evaluation setups probing perception, reasoning, and behavioral reliability under uncertainty. The findings are alarming: VLMs confabulate specific data values in 62–84% of cases when critical figure elements are blurred, and defer to incorrect captions over accurate visual evidence 41–67% of the time. Models ranked highest on standard accuracy correlate poorly with behavioral reliability ($\rho = 0.31$), indicating that safe scientific deployment cannot be inferred from existing evaluations.

AT A GLANCE

Problem: Standard VLM benchmarks measure accuracy on clean figures but not behavioral reliability when visual evidence is missing or misleading — a critical gap for scientific AI assistants.

Why it matters: Scientific AI tools are being deployed for literature review and data extraction; a model that fabricates values for blurred axes or trusts wrong captions poses real risks.

Method: 250 expert-annotated figures extended via image transformations, resistance probes, caption-bias probes, and selective-blur targets, producing 34,000+ evaluation setups.

Result: VLMs confabulate on 62–84% of blur conditions; caption-bias rate 41–67%; standard accuracy rank correlates poorly with behavioral reliability ($\rho = 0.31$).

Evidence: Resistance probe failures (71% compliance with false premises) confirm that models do not systematically detect their own visual limitations.

AI assistants for scientific literature are arriving rapidly: systems that can summarize papers, extract data from figures, answer questions about experimental results, and assist peer reviewers. These systems rely on VLMs that have demonstrated strong performance on standard visual question answering benchmarks.

Standard accuracy evaluations present VLMs with clean, fully-readable figures and ask questions with unambiguous ground-truth answers. This measures one important capability — accurate perception and reasoning — but ignores a second capability that is equally important for scientific use: *knowing what you cannot see*.

When a research figure has a blurred axis label, a partially occluded data series, or a misleading caption inserted during an adversarial or unintentional editing step, a reliable scientific AI assistant should express appropriate uncertainty. Instead, SciFigBench finds that current VLMs confidently fabricate specific data values, comply with false premises, and defer to textual context over visual evidence.

Why Existing Approaches Fall Short

Standard VQA and figure understanding benchmarks evaluate on clean images with unambiguous ground truth. They do not assess:

- Whether models appropriately express uncertainty when evidence is missing (rather than guessing)
- Whether models are biased by captions that contradict visual content
- Whether models detect fine-grained degradation of specific visual elements (e.g., one axis blurred while another is intact)

Hallucination benchmarks typically test factual grounding in natural images (“is there a cat in the image?”) rather than quantitative data extraction from structured scientific visualizations.

SciFigBench fills the gap by systematically varying the *state* of the visual evidence rather than just the questions asked about it.

Core Mechanism

Base figures: 250 scientific figures collected from STEM research papers across biology, physics, chemistry, computer science, and medicine. Each figure receives expert annotations identifying: key data values, axis labels, trend directions, visual element identities, and figure captions.

Evaluation setups (>34,000 total):

- (1) **Image transformations:** Gaussian blur, label removal, color-region masking applied to specific ele-

The Big Picture

ments while leaving others intact.

- (2) **Reasoning questions:** Quantitative (“What is the value at $x = 3$?”) and qualitative (“Which series shows the steeper slope?”).
- (3) **Resistance probes:** Model is told (falsely) that a specific value can be read from a blurred region — tests whether it complies or refuses.
- (4) **Caption-bias probes:** Figure paired with a caption that states a trend opposite to what the graph shows — tests whether model trusts visual evidence or caption.
- (5) **Selective-blur targets:** Single data elements blurred while the rest of the figure is intact — tests fine-grained awareness of local information availability.

Technical Formulation

The **behavioral reliability score (BRS)** for model M on perturbation condition set $\mathcal{P}$:

$$\text{BRS}(M) = \frac{1}{|\mathcal{P}|} \sum_{c \in \mathcal{P}} \frac{1}{|Q_c|} \sum_{q \in Q_c} \mathbf{1}[M(f_c, q) \text{ is behaviorally appropriate}]$$

where f_c is figure f under perturbation c , Q_c is the question set for condition c , and “behaviorally appropriate” means the model expresses uncertainty or refuses when evidence is unavailable, and trusts visual evidence over conflicting captions.

The **caption-bias score (CBS)**:

$$\text{CBS}(M) = \mathbb{E}_{(f,q)} [\mathbf{1}[M(f, q, \text{cap}_{\text{wrong}}) \neq M(f, q, \text{no cap})]]$$

where $\text{cap}_{\text{wrong}}$ is a caption that contradicts visual truth.

The **refusal rate (RR)** on resistance probes:

$$\text{RR}(M) = \mathbb{E}_{(f,q,\text{probe})} [\mathbf{1}[M \text{ declines to provide value}]]$$

Learning or Inference Procedure

SciFigBench is an evaluation benchmark with no training component. Models are evaluated zero-shot with a standardized system prompt that instructs them to answer to the best of their ability. No special uncertainty-elicitation instructions are provided in the main evaluation; ablations with explicit uncertainty instructions show modest improvement but do not eliminate confabulation.

What the Guarantee Says

No formal guarantee is provided. Ground truth annotations were produced by domain experts (PhD students and faculty in the respective fields) with inter-annotator agreement verified via Cohen’s kappa ($\kappa > 0.82$ across all annotation types). Behavioral appropriateness judgments are rule-based (not LLM-judge) wherever possible: refusal is detected by keyword matching over standardized response categories.

Experimental Findings

- **Confabulation under blur:** VLMs provide specific numerical answers 62–84% of the time when the asked-about visual element is blurred, nearly unchanged from clean conditions.
- **Caption bias:** Models align with the contradictory caption over correct visual evidence 41–67% of the time. Larger models show more bias, not less.
- **Resistance probe compliance:** Models comply with false premises (generate values from blurred regions) 71% of the time. Refusal rate is below 15% for all evaluated models.
- **Rank-reliability correlation:** Spearman $\rho = 0.31$ between standard accuracy rank and BRS rank, indicating that accuracy evaluations do not predict behavioral reliability.
- **Domain variation:** Biology and chemistry figures show higher caption-bias rates than computer science figures, possibly due to differences in training data figure-caption co-occurrence.

Ablations and Interpretation

Uncertainty elicitation: Prompting models to “express uncertainty when you cannot read a value” reduces confabulation from 74% to 52% — improvement, but not elimination. Models still confabulate more than half the time despite explicit instruction.

Blur severity: Confabulation rate is surprisingly stable across blur intensities from mild ($\sigma = 2$) to severe ($\sigma = 20$). This suggests models are generating values from semantic expectations rather than partially visible pixel information.

Caption-bias by model size: The largest evaluated models (7B–70B parameters) exhibit more caption bias than smaller models (1B–3B), consistent with larger models placing more weight on linguistic context relative to visual evidence during generation.

Selective vs. global blur: When only a single axis label is blurred while the rest of the figure is intact, confabulation rate (78%) slightly exceeds the global-blur rate (74%), suggesting models do not notice fine-grained local information loss.

Reference

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